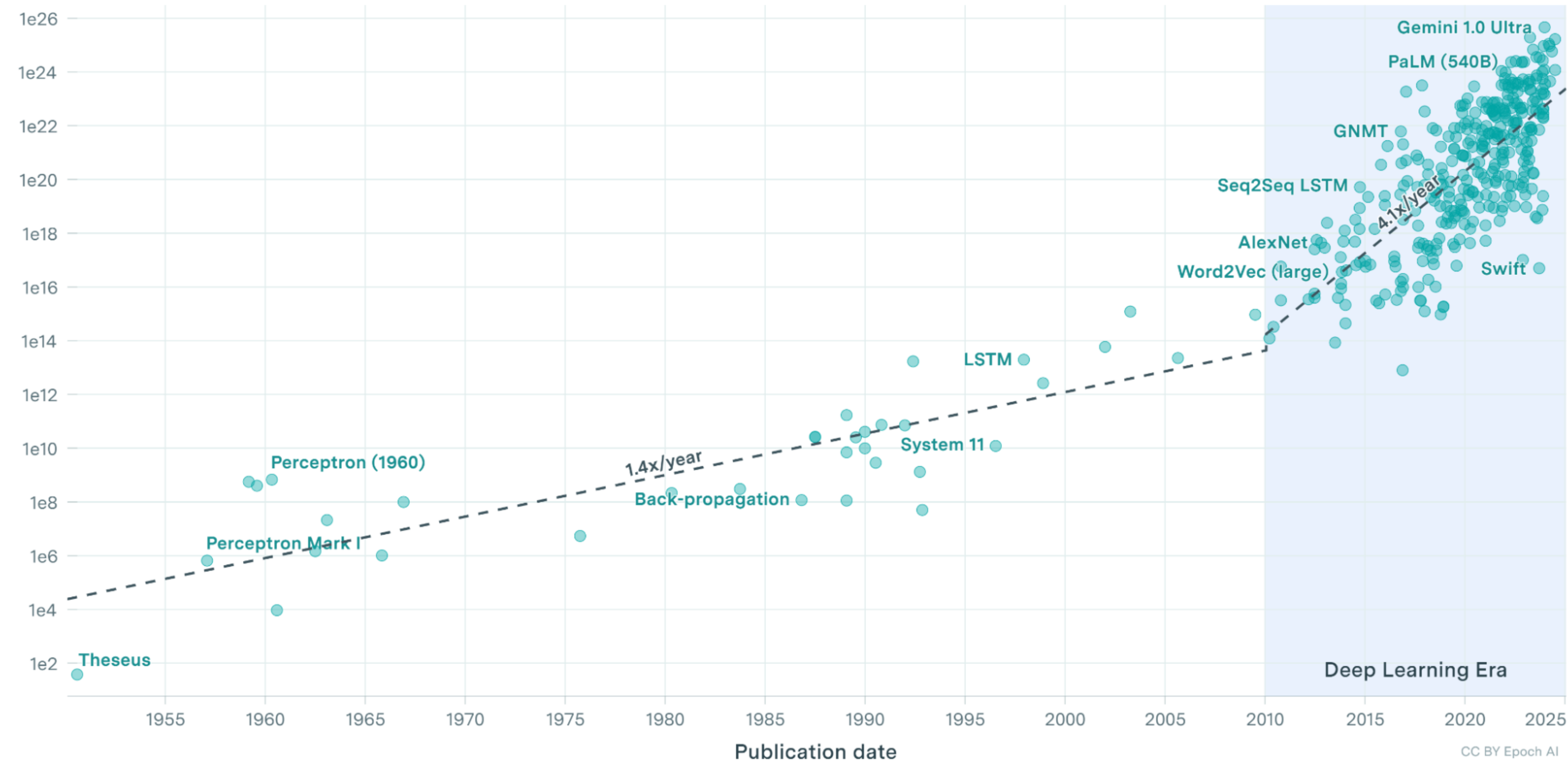


Notable AI Models

Training compute (FLOP)



ImageNet Classification with Deep Convolutional Neural Networks

Alex Krizhevsky
University of Toronto
kriz@cs.utoronto.ca

Ilya Sutskever
University of Toronto
ilya@cs.utoronto.ca

Geoffrey E. Hinton
University of Toronto
hinton@cs.utoronto.ca

Abstract

We trained a large, deep convolutional neural network to classify the 1.2 million high-resolution images in the ImageNet LSVRC-2010 contest into the 1000 different classes. On the test data, we achieved top-1 and top-5 error rates of 37.5% and 17.0% which is considerably better than the previous state-of-the-art. The neural network, which has 60 million parameters and 650,000 neurons, consists of five convolutional layers, some of which are followed by max-pooling layers

ADAM: A METHOD FOR STOCHASTIC OPTIMIZATION

Diederik P. Kingma*
University of Amsterdam, OpenAI
dpkingma@openai.com

Jimmy Lei Ba*
University of Toronto
jimmy@psi.utoronto.ca

ABSTRACT

We introduce *Adam*, an algorithm for first-order gradient-based optimization of stochastic objective functions, based on adaptive estimates of lower-order moments. The method is straightforward to implement, is computationally efficient, has little memory requirements, is invariant to diagonal rescaling of the gradients, and is well suited for problems that are large in terms of data and/or parameters. The method is also appropriate for non-stationary objectives and problems with very noisy and/or sparse gradients. The hyper-parameters have intuitive interpretations and typically require little tuning. Some connections to related algorithms, on which *Adam* was inspired, are discussed. We also analyze the theoretical convergence properties of the algorithm and provide a regret bound on the convergence rate that is comparable to the best known results under the online convex optimization framework. Empirical results demonstrate that Adam works well in practice and compares favorably to other stochastic optimization methods. Finally, we discuss *AdaMax*, a variant of *Adam* based on the infinity norm.

Attention Is All You Need

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
usz@google.com

Llion Jones*
Google Research
lleion@google.com

Aidan N. Gomez* †
University of Toronto
aidan@cs.toronto.edu

Lukasz Kaiser*
Google Brain
lukaszkaiser@google.com

Illia Polosukhin* ‡
illia.polosukhin@gmail.com

Abstract

inant sequence transduction models are based on complex recurrent or onal neural networks that include an encoder and a decoder. The best ig models also connect the encoder and decoder through an attention m. We propose a new simple network architecture, the Transformer, ily on attention mechanisms, dispensing with recurrence and convolutions Experiments on two machine translation tasks show these models to or in quality while being more parallelizable and requiring significantly to train. Our model achieves 28.4 BLEU on the WMT 2014 English- n translation task, improving over the existing best results, including s, by over 2 BLEU. On the WMT 2014 English-to-French translation task, l establishes a new single-model state-of-the-art BLEU score of 41.8 after or 3.5 days on eight GPUs, a small fraction of the training costs of the els from the literature. We show that the Transformer generalizes well to is by applying it successfully to English constituency parsing both with limited training data.

Observation of a New Particle in the Search for the Standard Model Higgs Boson with the ATLAS Detector at the LHC

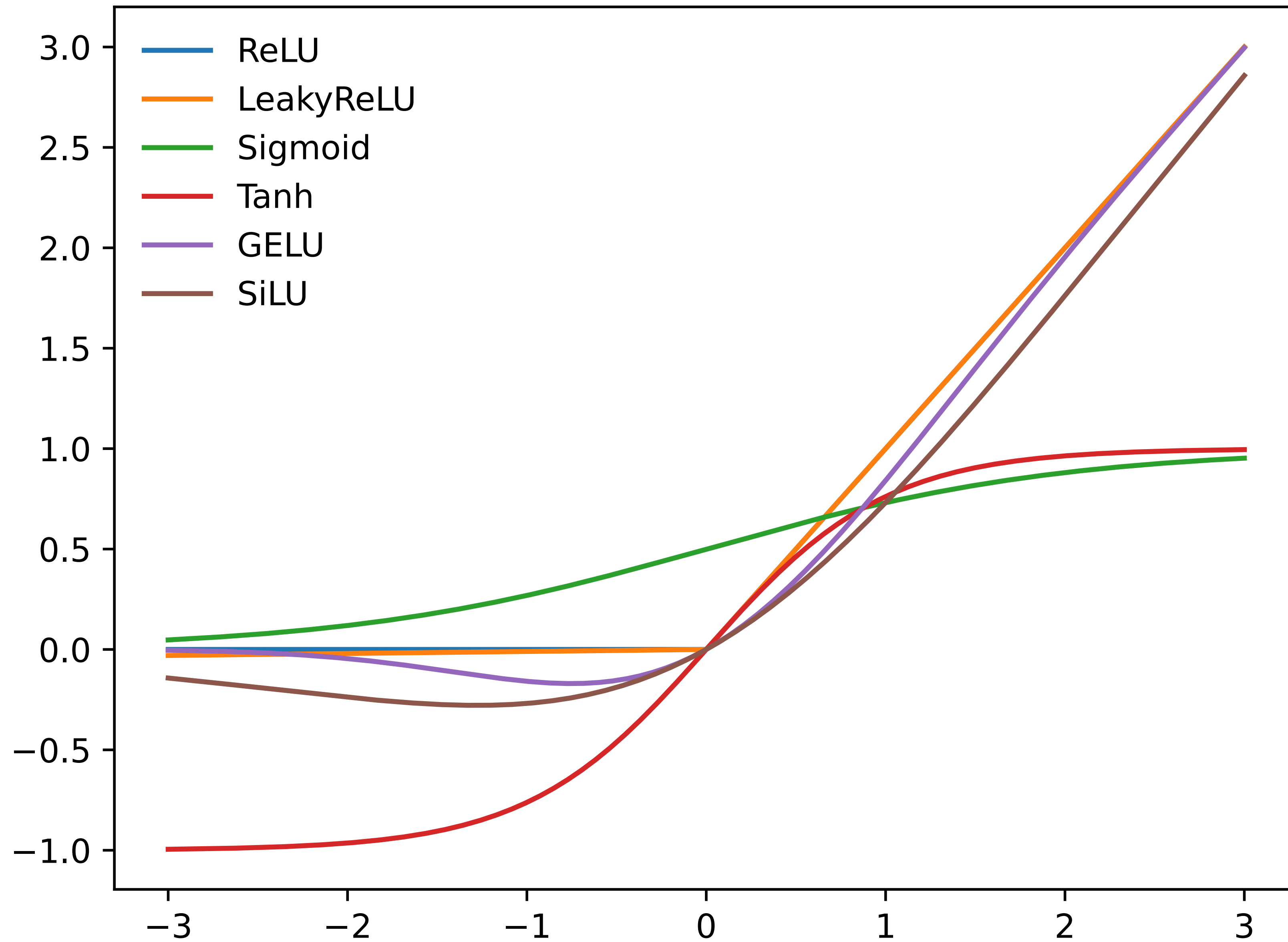
The ATLAS Collaboration

This paper is dedicated to the memory of our ATLAS colleagues who did not live to see the full impact and significance of their contributions to the experiment.

Abstract

A search for the Standard Model Higgs boson in proton-proton collisions with the ATLAS detector at the LHC is presented. The datasets used correspond to integrated luminosities of approximately 4.8 fb⁻¹ collected at $\sqrt{s} = 7$ TeV in 2011 and 5.8 fb⁻¹ at $\sqrt{s} = 8$ TeV in 2012. Individual searches in the channels $H \rightarrow ZZ^{(*)} \rightarrow 4\ell$, $H \rightarrow \gamma\gamma$ and $H \rightarrow WW^{(*)} \rightarrow e\nu\mu\nu$ in the 8 TeV data are combined with previously published results of searches for $H \rightarrow ZZ^{(*)}$, $WW^{(*)}$, $b\bar{b}$ and $\tau^+\tau^-$ in the 7 TeV data and results from improved analyses of the $H \rightarrow ZZ^{(*)} \rightarrow 4\ell$ and $H \rightarrow \gamma\gamma$ channels in the 7 TeV data. Clear evidence for the production of a neutral boson with a measured mass of 126.0 $\pm$ 0.4 (stat) $\pm$ 0.4 (sys) GeV is presented. This observation, which has a significance of 5.9 standard deviations, corresponding to a background fluctuation probability of 1.7×10^{-9} , is compatible with the production and decay of the Standard Model Higgs boson.

Popular activation functions



```

import torch
from torch.autograd import Function

class SquareFunction(Function):
    @staticmethod
    def forward(ctx, input):
        # Save what backward needs. Saving `input` (not `output`) because
        #  $d/dx (x^2) = 2x$ .
        ctx.save_for_backward(input)
        return input ** 2

    @staticmethod
    def backward(ctx, grad_output):
        (input,) = ctx.saved_tensors
        # Chain rule:  $dL/dx = dL/dy * dy/dx = grad\_output * 2x$ 
        grad_input = grad_output * 2 * input
        return grad_input # one return value per forward input

```

Momentum demo

distill.pub/2017/momentum

```
class torch.optim.adamw.AdamW(params, lr=0.001, betas=(0.9, 0.999), eps=1e-08, weight_decay=0.01, amsgrad=False, *,
maximize=False, foreach=None, capturable=False, differentiable=False, fused=None)
```

Implements AdamW algorithm, where weight decay does not accumulate in the momentum nor variance.

input : γ (lr), β_1, β_2 (betas), θ_0 (params), $f(\theta)$ (objective), ϵ (epsilon)
 λ (weight decay), *amsgrad*, *maximize*
initialize : $m_0 \leftarrow 0$ (first moment), $v_0 \leftarrow 0$ (second moment), $v_0^{max} \leftarrow 0$

for $t = 1$ to ... **do**

if *maximize* :

$g_t \leftarrow -\nabla_{\theta} f_t(\theta_{t-1})$

else

$g_t \leftarrow \nabla_{\theta} f_t(\theta_{t-1})$ ← Gradient $\nabla_{\theta} f_t(\theta_{t-1})$ from autograd

Weight decay λ → $\theta_t \leftarrow \theta_{t-1} - \gamma \lambda \theta_{t-1}$

$m_t \leftarrow \beta_1 m_{t-1} + (1 - \beta_1) g_t$

$v_t \leftarrow \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$

$\widehat{m}_t \leftarrow m_t / (1 - \beta_1^t)$

if *amsgrad*

$v_t^{max} \leftarrow \max(v_{t-1}^{max}, v_t)$

$\widehat{v}_t \leftarrow v_t^{max} / (1 - \beta_2^t)$

else

$\widehat{v}_t \leftarrow v_t / (1 - \beta_2^t)$

Update with learning rate γ → $\theta_t \leftarrow \theta_t - \gamma \widehat{m}_t / (\sqrt{\widehat{v}_t} + \epsilon)$

return θ_t

Link to coding exercises:

github.com/spinjo/ml-lecture-ipp-2026